V-M-81d
16-APR-1943
SUPERSEDING
Fed. Spec. V-M-81c
October 24, 1956
(See Section 6)

FEDERAL SPECIFICATION

MATTRESSES, BED, FELTED, COTTON

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the material and manufacture of felted cotton mattresses.

1.2 Classification.

1.2.1 *Types and grades.* Felted cotton mattresses shall be of the following types and grades, as specified (see 6.2):

Type I.—Plain box edge.
Type II.—Rolled box edge.
Type III.—Plain edge, not boxed.

Grade A.—100 percent cotton.
Grade B.—75 percent linters, 25 percent cotton.
Grade C.—60 percent linters, 40 percent cotton.

1.2.2 *Sizes.* Felted cotton mattresses shall conform to the sizes shown in table I, as specified (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378—Plastic Film (Polyethylene Thin Gage).
T-T-871—Twine, Cotton, Wrapping.
V-T-276—Thread, Cotton.
CCC-C-429—Cloth, Cotton, Osnaburg.
CCC-C-436—Cloth, Cotton, Ticking, Twill.
CCC-C-467—Cloth, Jute (or Kenaf), Burlap.
CCC-T-191—Textile Test Methods.
DDD-L-20—Label; for Clothing, Equipage, and Tentage (General Use).
DDD-T-86—Tape, Textile, Cotton, General Purpose (Gray, Bleached or Dyed).
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas

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City, Mo., Dallas, Denver, San Francisco, Seattle, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-B-3149—Baling of Clothing and Equipage.

MIL-T-20618—Treatment, Fire-Laundry-Dry-Cleaning and Mildew Resistant (for Cotton Fabrics).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MS-16686—Medical Services Symbol.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

U. S. Department of Agriculture Publication:

SRA-AMS-180—Cotton and Cotton Linters Standards.

(Application for copies should be addressed to the Department of Agriculture, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, the contractor shall submit to the contracting officer for approval a sample of the finished mattress conforming to the requirements of this specification.

3.2 Materials.

3.2.1 *Felt.* The felt shall be cleaned and well garnetted to produce unbroken laminated sheets. The use of oil in the fabrication of the felt shall not be permitted.

3.2.1.1 *Grade A.* Felt for grade A mattresses shall be made of 100 percent 1/2-inch staple cotton conforming to 3.2.1.4.

3.2.1.2 *Grade B.* Felt for grade B mattresses shall be made of a blend of 75 percent linters conforming to 3.2.1.5 and 25 percent 7/8-inch staple cotton conforming to 3.2.1.4 (see 6.3).

3.2.1.3 *Grade C.* Felt for grade C mattresses shall be made of a blend of 60 percent linters conforming to 3.2.1.5 and 40 percent 1/2-inch staple cotton conforming to 3.2.1.4.

3.2.1.4 *Cotton.* The cotton shall be new (not previously used) 7/8-inch staple cotton for grade B mattresses, and 1/2 inch staple cotton for grades A and C mattresses, not lower in grade than No. 7 or low middling, No. 7 or low middling spotted, No. 4 or strict middling gray, conforming to the standard of the United States Department of Agriculture Publication No. SRA-AMS-180. The use of reginned cotton will be permitted provided the reginned cotton is equivalent in grade to the standards in USDA Publication No. SRA-AMS-180 for cotton, No. 7 or low middling No. 7 or low middling spotted, or No. 4 or strict middling gray. Tinged or stained cotton shall not be used.

3.2.1.5 *Linters.* The linters for grades B and C mattresses shall be new (not previously used), conforming to grade 5, staple 5, of USDA Publication No. SRA-AMS-180.

3.2.2 *Cloth.*

3.2.2.1 *All types and grades except as specified in* 3.2.2.2. The cloth for mattress cases shall be ticking conforming to type II of CCC-C-436.

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3.2.2.2 *Type I, Grade B (Navy only).* The cloth for the mattress case shall be ticking conforming to type II of CCC-C-436, and in addition shall be treated for fire resistance in accordance with MIL-T-20618 except that the average char length shall be not more than 4.0 inches with a maximum char length of not more than 4.5 inches for any individual specimen. The requirements for after dry cleaning, mildew resistance and fastness to light shall not apply. The treated ticking shall show an air permeability of not less than 10.0 cubic feet per minute per square foot. The finished fabric shall be preshrunk so that the residual shrinkage shall not exceed 2.0 percent in either warp or filling directions. Testing shall be as specified in 4.3.1.1.

3.2.3 *Thread.* The thread shall be unbleached cotton thread ticket No. 30, 3 ply conforming to type IA1 of V-T-276.

3.2.4 *Twine for hand stitching.* The twine for hand stitching the sides, ends, and rolled edge shall be waxed or polished. Either cotton or flax twine having a breaking strength of not less than 20 pounds shall be used. Tests shall be conducted as specified in 4.3.1.1.

3.2.5 *Tufting twine, braid, or tape.* The twine or braid used in tufting shall have a breaking strength of not less than 32 pounds. Tufting tape shall be not more than 5/8 inch wide, and shall have a breaking strength of not less than 50 pounds. Testing shall be as specified in 4.3.1.1.

3.2.6 *Tape with metal locking strip.* The tape shall be a single 5/8-inch cotton tape conforming to class 1 of DDD-T-86. The metal locking strip shall be made from a corrosion resistant steel, of a type commercially available and commonly used in tufting mattresses.

3.2.7 *Labels.* All labels shall be sewn into the same edge seam at either end of each mattress.

3.2.7.1 *Statutory labels.* Statutory labels shall conform to the legal requirements of the state in which the mattresses are manufactured.

3.2.7.2 *Identification labels.* Each mattress shall have an identification label conforming to type I, class 5 of DDD-L-20.

3.2.7.3 *Medical label.* When specified (see 6.2), mattresses procured for medical activities shall have a label conforming to type I, class 9 of DDD-L-20, bearing manufacturer's name or registered trademark and a caduceus approximately 1 inch in height entered between letters "U" and "S" of appropriate size conforming to type I of MS-16686.

3.3 Construction.

3.3.1 *Mattress case.* The top and bottom sections of the mattress case shall each be made up of not more than two lengthwise pieces of cotton ticking specified in 3.2.2 properly matched and sewn together. Cross seams will not be permitted. The blue stripe of the ticking shall run lengthwise to the mattress.

3.3.1.1 *Edges and corners.* The box edges for types I and II mattresses shall be formed of not more than four separate pieces. When more than one piece is used in constructing the box edge, the pieces shall be of sufficient length to allow joining seams to be at a corner of the mattress and the pattern shall be properly matched. The corners for types I and III mattresses shall be of the square type. Corners on type II mattresses shall be slightly rounded to permit the use of a roll edge machine. The distance between the finished seams forming the box edge on types I and II mattresses shall be 2-inches ± 1/2 inch less than the thickness of the mattress when measured as specified in 4.3.3.2.

3.3.2 *Machine stitching and seaming.*

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3.3.2.1 *Stitches.* Stitching shall conform to stitch type 301 or 401 of Fed. Std. No. 751 with 8 to 12 stitches per inch.

3.3.2.2 *Seams.* When two pieces of ticking are used (see 3.3.1), they shall be stitched with stitch type 401 and seamed with seam type LSb-2 or stitch type 301 and seam type SSa-1, the box ends shall be seamed closed using seam type SSc-1 conforming to Fed. Std. No. 751.

3.3.2.3 Thread breaks, or any end of stitching not caught in other seams or stitchings, shall be backstitched or backtacked not less than 1 inch. All thread ends shall be trimmed.

3.4 **Felt distribution.** From 80 to 90 percent of the felt batts specified in 3.2.1 shall be distributed in layers of unbroken laminated sheets to produce uniform thickness. The remainder of the felt shall be uniformly and equally distributed over the middle one-third of the length of each face and the full width of the mattress to provide crowning.

3.5 **Tufting.** Mattresses shall be tufted on both sides using either tufting twine, braid, or tape specified in 3.2.5, or tape with a corrosion resistant metal locking strip specified in 3.2.6, fastened to each end. The twine, braid or tape tying tufts shall be drawn through both sides of the mattress, to a length not more than the thickness of the mattress minus 2 inches and tied. The arrangement of the tufts shall be in a diamond pattern with spacing not more than 8 inches apart over the entire top and bottom surfaces. The number of tufts per mattress shall be as specified in table I.

3.6 **Size dimensions, weight, tufts.** Unless otherwise specified (see 6.2) dimensions, weight and tufts shall conform to table I for each mattress.

3.7 **Hand stitching (sides and ends).** The mattress cover and felt along sides and ends of mattress shall be hand sewn with two rows of single stitches sewn through the boxing along each side and each end. The two rows of stitches shall be equally spaced between the edges forming the boxing and the individual stitches and shall be evenly spaced and drawn. The number of stitches shall conform to table I. For mattresses not tabulated in table I, the number of stitches required shall be interpolated from table I.

3.8 **Rolled edges (type II only).** The rolled edges shall be made uniform in size and well filled with felt to form a compact, even roll. Stitching shall be 2 inches or less per stitch.

3.9 **Workmanship.** The finished mattresses shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Contractor's inspection responsibility.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Pre-acceptance inspection.** The preproduction sample submitted in accordance with 3.1 shall be visually inspected and tested in accordance with 4.3.2.1.1 thru 4.3.2.1.3.

4.3 **Inspection for acceptance.** Except when otherwise specified by Civilian Government Activities or otherwise indicated, inspection shall be in accordance with the provisions of MIL-STD-105.

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TABLE I.- Dimensions, weight, tufts, and hand stitching

Dimensions			Weight (minimum)	Tufts (minimum)	Hand Stitching Stitches per single row (minimum)	
Length ± 1/4 inch	Width ± 1/4 inch	Thickness ± 1/4 inch			on each side	on each end
Inches	Inches	Inches	Pounds			
72-1/2	26	4	18	28	47	20
76	26	3	15	28	49	20
73	27	4	28	28	47	20
74	28	4	25	28	48	20
77	28	4-1/2	23-1/2	28	50	20
76	30	4	23	28	49	21
78	31	4	27	38	51	22
75	33	6	37	38	49	23
76	34	4	26	38	49	24
78	36	4-1/2	31	38	51	26
75	38	6	42	38	49	27
76	39	4-1/2	38	38	49	28

4.3.1 *Inspection of components.*

4.3.1.1 *Testing of components.* In addition to the quality assurance testing provisions of the subsidiary specifications, testing shall be performed on components and materials listed in table II for the characteristics shown therein. The inspection level shall be L-1. The AQL for sample unit requirements shall be 6.5 defects per 100 units. No AQL is applicable for lot average requirements. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in the following table, they shall be as required by the referenced test method.

4.3.1.2 *Felt.* Unless otherwise specified (see 6.2), the cotton or linters, or both shall be examined to determine conformance with 3.2.1 through 3.2.1.3 inclusive, by classifiers licensed by the U. S. Department of Agriculture; or representative samples each weighing not less than 1/2 pound, of the cotton or linters, or both, and of the felt may be forwarded to the administrative office concerned at Washington, D. C., where inspection by the proper board of the U. S. Department of Agriculture will be obtained. The opinion of this board as to the true grade or quality will be final.

4.3.1.3 *Certification of compliance.* Components and materials listed below may

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TABLE II.- Test methods of components

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	Number of determinations per sample unit	Results reported as	Sample unit
Linters as applicable (1 bale)	Grade	3.2.1.5	Visual	-	X	1	Pass or fail	1/2 lb.
	Staple	3.2.1.5	Visual	-	X	1	Pass or fail	1/2 lb.
Cotton as applicable (1 bale)	Grade	3.2.1.4	Visual	-	X	1	Pass or fail	1/2 lb.
	Staple	3.2.1.4	Visual	-	X	1	Pass or fail	1/2 lb.
Ticking (1 yard full width)	Air permeability 1/	3.2.2.2	5450 CCC-T-191	X	-	-	-	2 yards full width
	Shrinkage 2/	3.2.2.2	5550 CCC-T-191	X	-	-	-	2 yards full width
Twine (hand stitched) (1 cone or spool)	Material identification	3.2.4	1200 or 1210 as applicable CCC-T-191	X	-	-	-	1 cone or spool
	Breaking strength	3.2.4	4102 CCC-T-191	X	-	-	-	1 cone or spool
Tufting twine or braid (1 cone or spool)	Breaking strength	3.2.5	4102 CCC-T-191	X	-	-	-	1 cone or spool
Tufting tape	Breaking strength	3.2.5	5100 3/ CCC-T-191	X	-	-	-	2 yards
	Width	3.2.5	Std Comm'l	X	-	1	Pass or fail	

1/ The air permeability may be determined by other apparatus provided a suitable correlation is established with specified method and any corrections in results made as may be indicated.

2/ The three specimens required for the shrinkage test shall be selected from the cloth (sample unit) as follows: one specimen from each side of the cloth to within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both inclosed in parenthesis.

3/ Breaking strength shall be determined on the full width of the tape and the jaw dimensions shall be greater than the width of the tape.

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be accepted on the basis of a contractor's certification of compliance for requirements specified in applicable paragraphs of this specification.

Component	Characteristic	Requirement paragraph
Felt	Oil not used and shall be well garnetted to produce unbroken laminated sheets having applicable composition and distribution.	3.2.1 (3.2.1.1, 3.2.1.2 or 3.2.1.3, as applicable) and 3.4.
Metal locking strip ..	Corrosion - resistant steel.	3.2.6.

4.3.1.4 *In-process inspection.* Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated non-conformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirements.

Operation or assembly	Requirement paragraph
Processing of felt	3.2.1
Composition (blending, when applicable) of felt	3.2.1.1, 3.2.1.2, or 3.2.1.3 (as applicable)
Distribution of felt	3.4

4.3.2 *Inspection of the end item.*

4.3.2.1 *Visual examination of mattresses.* The defects found during examination shall be classified in accordance with 4.3.2.1.1, 4.3.2.1.2, 4.3.2.1.3, and 4.3.2.1.4. The lots shall be inspected in accordance with 4.1.5. The sample unit for these examinations shall be one mattress (as applicable).

4.3.2.1.1 *Examination of mattress for defects in design, material, construction and workmanship.*

Examine	Defect	Classification	
		Major	Minor
Quality of ticking	Any hole, cut or tear (except underneath tuft), or where twine forms rolled edge	X	
	Any weak place seriously affecting serviceability	X	
	Any weak place affecting serviceability but not seriously		X
	Any objectionable odor	X	
	Not fabric specified	X	
	Multiple floats or skips 1/2 inch or more in combined directions	X	
	Multiple floats or skips less than 1/2 inch in combined directions		X
	Three or more contiguous broken or missing warp or filling threads	X	
	Two contiguous broken or missing warp or filling threads		X
	Loose and weakening slub, 1/8 inch or more in thickness	X	
	Spot or stain 1/4 inch or over in diameter, clearly noticeable when viewed at a distance of 6 feet ..		X
Seams and stitching	Any required seam or stitching operation omitted	X	
	Mattress case (other than closing end) not seamed with seam type SSa-1 or LSb-2		X

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Examine	Defect	Classification	
		Major	Minor
Seams and stitching (cont'd)	Closing end not seamed with seam type SSc-1		X
	Not stitch type 301 or 401		X
	Five to seven stitches per inch ..		X
	Less than five stitches per inch ..	X	
	More than twelve stitches per inch ..		X
	Ends of any seam backstitched or back-tacked less than 1 inch (except when end of seam is caught in another seam or stitching)		X
	Thread breaks not backstitched or back-tacked less than 1 inch		X
	Raw edge more than 1 inch in length ..		X
	Open seam more than 1 inch in length	X	
	Open seam 1 inch or less in length		X
	Note: A seam shall be classified as open when one or more stitches are broken, or when two or more continuous skipped stitches or run-offs occur.		
	Stitch tension loose, resulting in a loosely secured seam		X
	Stitch tension tight (stitches break when normal strain is applied to the seam or stitching)		X
	Note: Puckering is evident of tight tension, when puckering is evident, the inspector shall test seams by exerting pull on seam.		
Seams and stitching (cont'd)	Needle chew likely to result in a hole or tear	X	
	Seam badly puckered, pleated, twisted, or caught in unrelated operation		X
	Thread ends not trimmed throughout		X
Construction and workmanship	Not type specified .	X	
	Sagging appearance, i.e., filling not evenly distributed	X	
	Lumps clearly noticeable	X	
	Thin hollow places clearly noticeable .	X	
	Top section of mattress case fabricated of more than two lengthwise pieces		X
	Bottom section of mattress case fabricated of more than two lengthwise pieces		X
	Blue stripe of ticking does not run in lengthwise direction		X
	Top and bottom sections of mattress case not properly matched when sewn together, affecting appearance		X
	Any cross seam in top and bottom sections		X
Edges	Not constructed as specified	X	
	Box edge (when applicable), if separate, formed of more than four separate pieces ...		X

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Examine	Defect	Classification	
		Major	Minor
Edges (cont'd)	Joining seams (when applicable), of box edging (when applicable), not located at a corner		X
	Pattern not properly matched when joining seams		X
	Corners of the type I and type III mattresses not the square type		X
	Corners of the type II mattress not slightly rounded		X
Tufting	Material used in tufting not type specified		X
	Three or more tufts insecurely knotted or fastened	X	
	Two tufts insecurely knotted or fastened		X
	Not accomplished as specified		X
	Tufts not arranged in a diamond pattern		X
	Two or more less than number of tufts specified	X	
	One less than number of tufts specified		X
	Any knot or fastening device appearing above surface on either side of mattress		X
	Tear over 1/4 inch in ticking under tuft (check a minimum of 3 tufts on each side of mattress		X
	Spacing of tufts not sufficiently far apart to hold securely without ripping or tearing the intervening cloth		X
	Uneven overall tufting causing mattress to lean over or to have an uneven surface appearance	X	
Rolled edges (Type II only)	Hole, cut or tear under twine over 1/4 inch	X	
	Rolled edges not uniformly rolled affecting appearance		X
	Rolled edges not well filled to form a compact even roll		X
	Stitching more than 2 inches per stitch		X
	Any edge not rolled	X	
	Rolled edge stitching not single stitches		X
Hand stitching of side and end boxing	Rows of single stitches not equally spaced between edges of boxing, affecting appearance		X
	More than one broken stitch		X
	One broken stitch	X	
	Five or more stitches less than the minimum specified on side boxing (as applicable)	X	
	One, two, three, or four stitches less than the minimum number specified on side boxing (as applicable)		X
	Three or more stitches less than the minimum number specified on end boxing (as applicable)	X	
	One or two stitches less than the minimum number specified on end boxing (as applicable)		X
Labels	Missing, incomplete, incorrect or illegible		X

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4.3.2.1.2 *Examination of mattress for defects in dimensions.*

Examine	Defect
Length (see paragraph 4.3.3.3)	Average length is beyond tolerances indicated in table I.
Width (see paragraph 4.3.3.3)	Average width is beyond tolerances indicated in table I.
Spacing of tufts (both surfaces)	More than 8 inches apart.

4.3.2.1.3 *Examination of mattress for defects in weight.*

Examine	Defect
Weight (see 4.3.3.1)	Less than the minimum specified in table I.

4.3.2.1.4 *Examination of preparation for delivery requirements.* An examination shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete application of tape, incomplete closure of container flaps, or cloth covering, loose strapping, inadequate stapling, insecure sewing.
	Bulging or distortion of containers.
Content	Number per container is more or less than required.

4.3.2.1.5 *Inspection levels and AQL's.* The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

For examination in	Inspection level	AQL
4.3.2.1.1	I	Major 2.5
4.3.2.1.1	I	Total 6.5
4.3.2.1.2	L-6	1 Class 4.0
4.3.2.1.3	L-6	1 Class 4.0
4.3.2.1.4	L-4	1 Class 4.0

4.3.2.2 *Testing of the end item.* Testing of the completely fabricated mattress shall be performed in accordance with table III for the characteristics shown therein. The sample unit shall be one mattress, completely fabricated.

4.3.3 *Methods of measurement and weight.*

4.3.3.1 *Test for weight.* Weight may be determined under prevailing atmospheric conditions, except in the settlement of disputes. The weight of the mattress shall then be determined after having been brought to equilibrium with an atmosphere having a relative humidity of 65 ± 2 percent and a temperature of from 70° to 80° F. The material shall be considered to be in equilibrium when it shows no progressive change in weight.

4.3.3.2 *Measurement of thickness.* Thickness shall be taken as the average of measurement at each of the four corners of the mattress made by determining the free height under the corner of an 18-inch square panel with a total weight load of 4 pounds, and resting on the mattress, with two edges over the adjacent edges of the mattress.

4.3.3.3 *Measurement of length and width.* Length and width shall be taken as the average of three measurements in each direction. Before measuring, mattresses shall be shaken out after removal from the bales and laid in a flat position.

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TABLE III.-

Characteristic	Requirement paragraph	Test method	Requirement applicable to		Number of determinations per sample unit	Results reported as	Inspection level	AQL
			Sample unit	Lot average				
Thickness (as applicable)	Table I	Paragraph 4.3.3.2	-	X	4	Average or 4 determinations to nearest 1/16 inch	L-1	-
Drawn * length of tuft	3.5	Visual	-	X	3	Average of 3 determinations to nearest 1/16 inch	L-1	-
Distance between seams forming box edge (Types I and II mattress)	3.3.1.1	Visual	-	X	3	Average of 3 determinations to nearest 1/16 inch	L-1	-

Note: *Cut tuft at knot and pull remaining portion of tuft from opposite side, measure the length of the tufting material to the remaining knot. Cut a total of three tufts from each mattress in the sample unit, measure the lengths and average the values found. After the determinations have been made, require the contractor to hand tuft the mattress where the tufts have been removed.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.* Each mattress shall be individually packaged in a snug-fitting polyethylene bag. The polyethylene flexible film shall be 0.004 inch thick conforming to type I, grade A, B, or C, finish 1 of L-P-378. All seams and closures shall be effected by heat sealing and shall conform to the heat seal strength requirement in L-P-378. Prior to or during the final heat sealing operation, excess air within the bag shall be expelled.

5.1.2 *Level C.* Mattresses shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Three mattresses of one type, grade and size only, packaged as specified in 5.1, shall be packed flat and wrapped in cotton osnaburg cloth conforming to type I, class 2, finish A of CCC-C-429; or jute (or kenaf) burlap cloth conforming to class 1 of CCC-C-467. The cloth covering shall be sewn with 16-ply cotton

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twine conforming to T-T-871, with approximately one stitch per inch and every third stitch knotted. Five inches of surplus covering shall be gathered on each of the four corners and securely sewn into ears for handles. Alternatively, the materials specified may be obtained in bias sewn tubing form as specified in MIL-B-3149. The tubing may be machine closed using stitch type 101 or 401 with 3 to 6 stitches per inch. Thread cotton machine ticket No. 12, 4-ply shall be used in the needle and ticket No. 16, 4-ply in the looper of the 401 stitch. The thread shall be in accordance with type IA1 of V-T-276.

5.2.1.1 When specified (see 6.2) three mattresses of one type, grade and size only, packaged as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to class 2, V3c or V3s, style FTC of PPP-B-636, except that the limensional and weight limitations shall be waived. Each shipping container shall be closed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 *Level B.* Mattresses shall be packed as required in 5.2.1.

5.2.2.1 When specified, three mattresses of one type, grade and size only, packaged as specified (see 6.2) shall be packed flat in a snug-fitting fiberboard shipping container conforming to type I or II, class 1, style FTC of PPP-B-636, except that the dimensional limitation shall be waived. Each container shall be reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.3 *Level C.* Mattresses, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military requirements.* In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129. The following marking shall appear in capital letters not less than 1-1/2 inches in height on at least two sides of each shipping container or cloth bag.

"STOW FLAT—DO NOT STOW ON END"

5.3.2.1 *Polyethylene bagged packages.* Polyethylene bagged packages shall have the required information legibly printed or stamped in black on a white paper label inserted within the bag at one end of the mattress so as to permit ready identification. The label shall indicate the following information:

STOCK NUMBER
NOMENCLATURE
TYPE
SIZE
GRADE
QUANTITY
CONTRACT NUMBER

6. NOTES

6.1 Intended use. These mattresses are intended for use by Civilian and Military agencies of the Government (see 6.3).

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

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(*b*) Type, grade, and size mattress required (1.2 and 3.6).

(*c*) When cotton and linters or both shall be examined (4.3.1.2).

(*d*) Whether Medical label is required (3.2.7.3).

(*e*) Selection of applicable levels of packaging and packing (5.1 and 5.2).

(*f*) When packing of mattresses in fiberboard boxes is required (5.2.1.1 and 5.2.2.1).

(*g*) Whether preproduction sample is required (see 3.1).

6.3 Supersession. This specification includes the requirements of MIL-M-16039B dated 11 October 1960, in part.

6.4 Items procured under this specification for military use are to be limited to the riety shown on the applicable military andards. Personnel of the Military Departments are requested to refer to these documents for guidance.

6.5 Transportation description. The transportation description and minimum weights applicable to this commodity are:

Rail:

Mattresses
Carload minimum weights:
In cars 40'9" or less in length, inside measurements — 10,000; 12,000; 15,000; 18,000; 22,000 and 30,000 pounds.
In cars exceeding 40'9" in length, inside measurements — 16,000; 18,000; 21,000; 24,000; 30,000 and 40,000 pounds.

Motor:
Mattresses
Truckload minimum weights:
10,000; 12,000; 15,000; 18,000; 22,000 and 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.6 Level A packaging and level B packing shall be used by ordering agencies only when shipment is known to be made to holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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